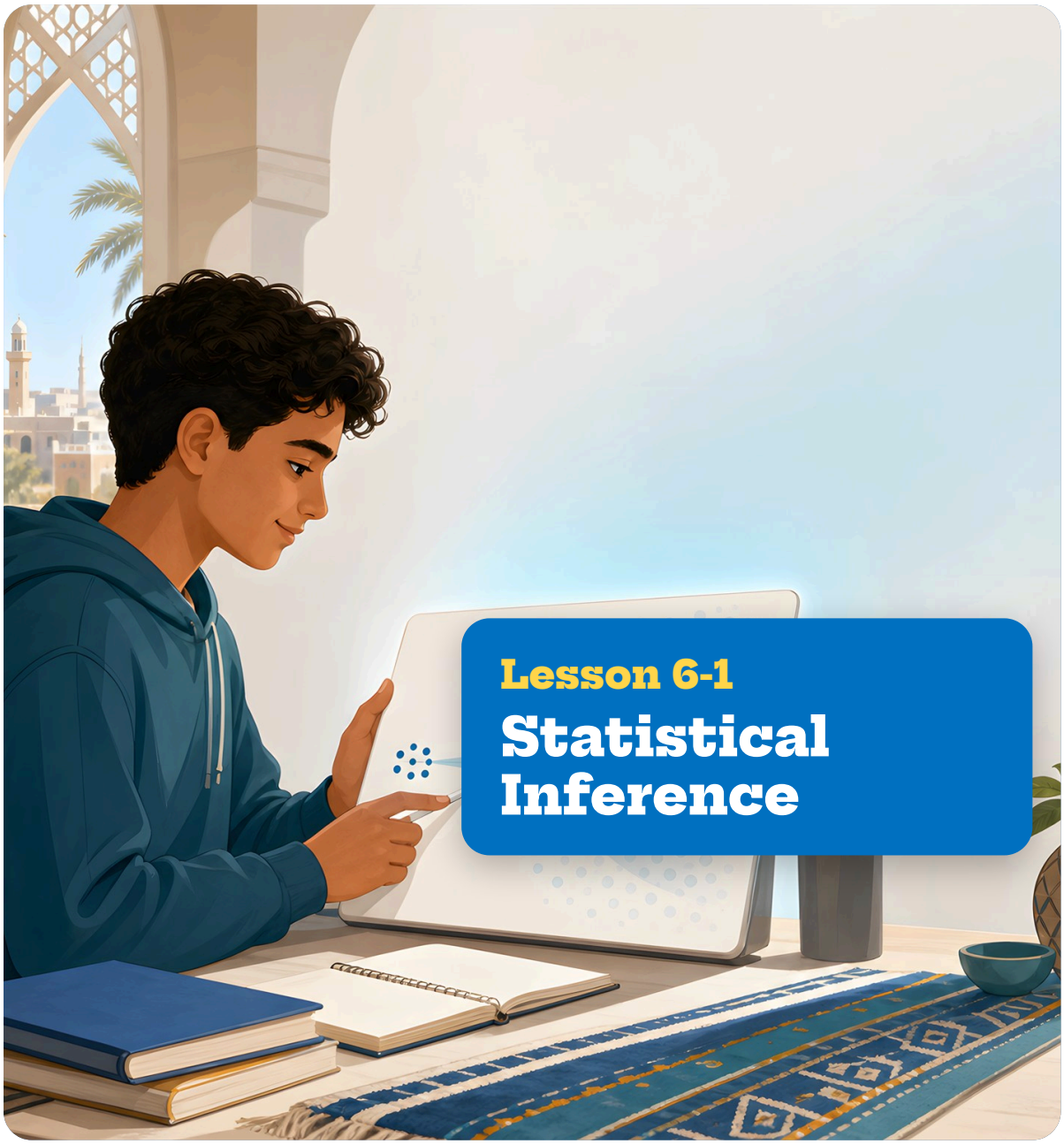


Learning Objectives

- Explain the difference between point estimation and interval estimation, including the meaning of a confidence level.
- Describe the procedure of hypothesis testing, and explain the cautions needed when interpreting its results.

You can never survey everyone, so you study a sample and reason back to the whole population — that is statistical inference. It answers two kinds of question: "about how big?" (estimation) and "is this real, or just chance?" (hypothesis testing). But because a sample is only part of the whole, every answer carries uncertainty. The skill is not to hide that uncertainty but to measure it — with confidence intervals, p-values, and significance levels — and to remember that "statistically significant" is not the same as "important."

? Guiding question: How can we reason from a sample to a whole population, and how do we measure the uncertainty in that reasoning?

📌 Point!

1. The Statistical Inference Stage

(1) A sample is drawn from the entire group that is the target of analysis (the population) and surveyed, and the properties of the population are then inferred from the results of the sample. Inferring the properties of the population from the sample in this way is called **(statistical inference)**.

(2) Statistical inference can be broadly divided into the following two approaches.

Approach	Description	Example
(Estimation)	Infer the properties of the population in numerical terms	Infer the population mean height from the sample mean height
(Hypothesis testing)	Determine whether a hypothesis about the population is correct	Determine whether a new learning method is more effective than the conventional one

(3) In either case, since the sample is only part of the population, inference always involves uncertainty. In statistical inference, judgment is made while considering the degree of that uncertainty.

2. Point Estimation and Interval Estimation

(1) **(Point estimation)**…estimating the properties of the population as a single value calculated from the sample. *e.g.,* If the sample mean height is 168 cm, then the population mean height is also estimated to be 168 cm.

(2) **(Interval estimation)**…estimating that the properties of the population fall within a certain range (interval). *e.g.,* The population mean height is estimated to be in the range of 166 cm to 170 cm.

(3) The interval used in interval estimation is called the **(confidence interval)**, and how often such intervals contain the population value over repeated surveys is called the **(confidence level)**. Levels of 95% or 99% are common.

Key Fact:
Statistical inference reasons from a sample to a population. Estimation gives a value (point) or a range (interval, with a confidence level); hypothesis testing weighs a null against an alternative using a p-value and a significance level. All of it carries uncertainty — and significance is not the same as practical importance.

Key Concepts:

- statistical inference
- point / interval estimation
- confidence interval & level
- null / alternative hypothesis
- p-value & significance level
- Type I / II errors

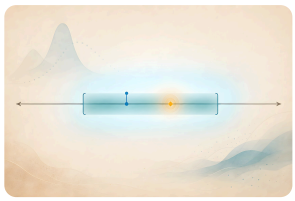
KEY TERMS
statistical inference — reasoning from a sample to the population.
estimation — inferring the population in numbers.
hypothesis testing — judging a claim about the population.

💡 PAUSE & THINK

A single value or a range — which estimate shows you how uncertain it is?

e.g., Suppose the population mean height is estimated by sample survey, and a confidence interval of “166 cm to 170 cm” is obtained at a 95% confidence level. The “95%” means that if the same survey method is repeated 100 times, about 95 of the resulting confidence intervals will contain the true population mean height.

(4) Point estimation is easy to understand because it is shown as a single value, but the uncertainty of the estimate is invisible. Interval estimation has width, so care is needed in interpreting the result, but it has the advantage of also showing the degree of uncertainty.



A point estimate is one value; an interval is a range.

3. The Idea of Hypothesis Testing

(1) Hypothesis testing…a procedure for determining whether a hypothesis about the population is acceptable based on the data at hand.

(2) In hypothesis testing, the following two hypotheses are set up.

Hypothesis	Description
(Null hypothesis) (H0)	A hypothesis serving as the starting point of the test, such as “no difference” or “no relationship”
(Alternative hypothesis) (H1)	The hypothesis adopted when the null hypothesis is rejected, such as “there is a difference” or “there is a relationship”

(3) In hypothesis testing, the probability that data collection could result in the data at hand is evaluated under the assumption that “there is no difference or relationship.” The reason for first assuming the hypothesis one wants to deny (the null hypothesis) is that, rather than directly showing “there is a difference or relationship,” it is logically easier to judge this by showing that, under the assumption “there is no difference or relationship,” the data at hand would be unlikely to be obtained. *e.g.,* It is difficult to directly show “the new learning method is effective.” Instead, one assumes “there is no effect” and then examines how low the probability is of obtaining a difference like the experimental result at hand, and if that probability is sufficiently small, one judges that “no effect” is unlikely.

(4) Based on the evaluation, the null hypothesis is judged to be either **(rejected (denied))** or **(not rejected)**.



Testing weighs evidence against the null hypothesis.

KEY TERMS
null hypothesis (H0) — the “no difference” starting point.
alternative hypothesis (H1) — adopted if H0 is rejected.
p-value — chance of data at least this extreme if H0 holds.

4. The Procedure of Hypothesis Testing

(1) Hypothesis testing follows this flow.

① Set the null hypothesis H0 and the alternative hypothesis H1

② Decide the **(significance level)** (5% = 0.05 and 1% = 0.01 are commonly used)

③ Compute the **(p-value)** from the data at hand

④ Compare the p-value with the significance level and make a judgment

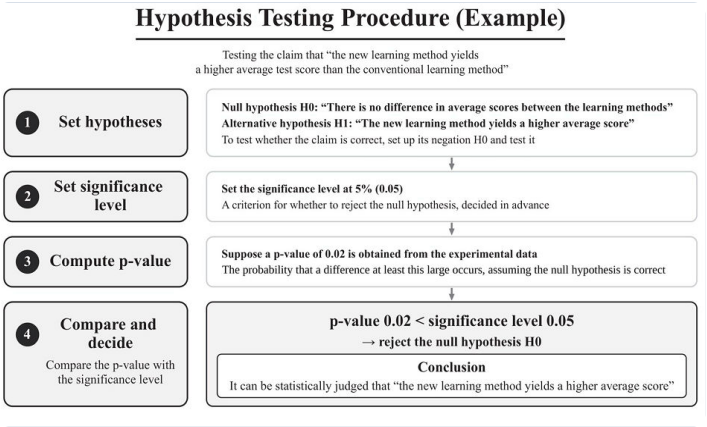
(2) p-value…assuming the null hypothesis is correct, the probability of obtaining a result as extreme as, or more extreme than, the data at hand. The

smaller the p-value, the less likely it is that the data would be obtained under the null hypothesis.

(3) Significance level…the threshold value used to decide whether to reject the null hypothesis.

(4) Criteria for judgment

Condition	Judgment
p-value < significance level	Reject the null hypothesis and adopt the alternative hypothesis (statistically significant)
p-value ≥ significance level	Do not reject the null hypothesis (cannot be said to be statistically significant)



The hypothesis testing procedure worked through an example

5. Major Hypothesis Testing Methods

(1) The test method used differs depending on the purpose of the analysis and the type of data. Major examples include the following.

Test	Main use	Example
(t-test)	Examine whether there is a difference in the means of two groups	Examine whether students' average test scores differ between a certain learning method and the conventional method
(Chi-squared test)	Examine whether there is a relationship between two categorical variables (data expressed by classification, such as gender, occupation, or preference)	Examine, from survey results, whether there is a relationship between gender and subject preference

6. Cautions Regarding Inference

(1) In statistical inference, there is always a possibility of judgment errors. There are two types of errors.

Type of error	Description
(Type I error)	Rejecting the null hypothesis when it is actually correct
(Type II error)	Not rejecting the null hypothesis when it is actually incorrect

PAUSE & THINK

If the p-value is 0.02 and the significance level is 0.05, do you reject the null or not?

PAUSE & THINK

Which test compares two group means, and which checks a relationship between categories?

KEY TERMS

Type I error — rejecting a true null hypothesis.
Type II error — keeping a false null hypothesis.
trade-off — lowering one error raises the other.

(2) Lowering the significance level reduces the Type I error, but tends to increase the Type II error. The two have a **(trade-off relationship)**.

(3) Care is needed in interpreting the p-value. The following misunderstandings must be avoided.

① The p-value is not “the probability that the null hypothesis is correct.”

② A small p-value does not mean “the effect is large.”

③ The comparison of the p-value with the significance level is not an absolute correctness criterion but a guide for judgment. The significance level (e.g., 0.05 or 0.01) is a value chosen according to the purpose of the analysis or the conventions of the field; falling below this value does not mean the result is certainly correct.

(4) Being statistically significant and being practically important are different. *e.g.,* Suppose an experiment compares two designs of a website, with click-rate data from 100,000 people split evenly. The old design is clicked by 0.4% of visitors, the new one by 0.5% — only 0.1 percentage points higher. Because the data set is large, this difference is likely to be statistically significant. However, whether that 0.1-point difference leads to actual profit or better user experience needs to be judged from a separate perspective.

(5) Statistical inference is a powerful means of drawing conclusions from data, but it is important to understand that it is a tool for judgment that includes uncertainty, and the results should be interpreted appropriately.



Type I is a false alarm; Type II is a miss.

PAUSE & THINK

With 100,000 people, a tiny difference can be “significant.” Does that make it important?

Worked Example

(1) Mark each statement (A–D) with ○ if true or × if false.

A Inferring the properties of the population from the sample is called statistical inference.

B Point estimation is a method that estimates the properties of the population within a certain range.

C In hypothesis testing, the probability that data collection could result in the data at hand is evaluated under the assumption that the null hypothesis is correct.

D If a result is statistically significant, the difference is also necessarily important in practice.

(2) Match each description (a–c) with the most appropriate term from the choices below (A–C).

a A hypothesis serving as the starting point of the test, such as “no difference” or “no relationship”

b The probability of obtaining a result as extreme as, or more extreme than, the data at hand, assuming the null hypothesis is correct

c The threshold value for deciding whether to reject the null hypothesis

[Choices] **A** p-value **B** Significance level **C** Null hypothesis

✓ Solution

(1)

A Correct definition of statistical inference. Therefore, ○.

B Estimating within a range is interval estimation. Point estimation estimates a single value. Therefore, ×.

C Correct description of the idea of hypothesis testing. Therefore, ○.

D Statistical significance and practical importance are different; with a large amount of data, even a tiny difference can become significant. Therefore, ×.

(2)

a: A hypothesis serving as the starting point of the test is the null hypothesis. **C**

b: The probability of obtaining the data under the null hypothesis is the p-value. **A**

c: The threshold value for rejection is the significance level. **B**

Try

1 Answer the following questions.

1. What is the term for inferring the properties of the population from the sample?
2. What is the term for the method of estimating that the properties of the population fall within a certain range?
3. What is the term for the interval used in interval estimation?
4. What is the term for a typical hypothesis testing method that examines whether there is a relationship between two categorical variables?

EXAM-STYLE QUESTION

Explain why researchers test a null hypothesis they hope to disprove, rather than proving the effect directly. **[6] Refer to:** what a small p-value shows.

2 Answer the following questions.

(1) From the following options (A–D), choose the one that most appropriately describes the procedure of hypothesis testing.

A Set the null and alternative hypotheses → decide the significance level → compute the p-value → make a judgment

B Compute the p-value → set the null hypothesis → decide the significance level → make a judgment

C Decide the significance level → adopt the alternative hypothesis → compute the p-value → set the null hypothesis

D Make a judgment → set the null hypothesis → decide the significance level → compute the p-value

(2) Suppose that, in a hypothesis test with the significance level set at 0.05 (5%), the p-value is 0.03. From the following options (A–D), choose the one that most appropriately describes the judgment in this case.

A Reject the null hypothesis and adopt the alternative hypothesis.

B Do not reject the null hypothesis.

C Reject the alternative hypothesis.

D Change the significance level to 0.01 and recalculate.

(3) For each of analysis purposes (a, b), choose the most appropriate test method from the choices below (A, B).

a You want to examine whether students' average test scores differ between two teaching methods.

b You want to examine whether there is a relationship between gender and the category of products purchased.

[Choices] **A** Chi-squared test **B** t-test

(4) From the following options (A–D), choose the one that most appropriately describes the p-value.

A The p-value is the probability that the null hypothesis is correct.

B The smaller the p-value, the larger the effect is.

C The p-value is the probability of obtaining a result as extreme as, or more extreme than, the data at hand, assuming the null hypothesis is correct.

D p-value < 0.05 is an absolute criterion of correctness.

Think as an Engineer — Investigate & Decide

A school club claims its new online quiz tool raises students' average score on weekly tests. You are asked to judge the claim fairly using a hypothesis test.

☐ **Trust the club's own summary** ☐ **Frame and test the claim properly**

PAUSE & THINK

p-value 0.03 against a 0.05 threshold — below or above? So what do you do?

KEY TERMS

significance level — the threshold for rejecting H0.
statistically significant — p below the threshold.
practically important — big enough to matter in real life.

1 · Frame the hypotheses. The club reports its tool group averaged 74 on the last quiz versus 70 for a control group of the same size. Write the null and alternative hypotheses for this exact claim in your own words, state what data you would still need to actually run the test (beyond these two means), and say which significance level you would set (0.05 or 0.01) and why.

2 · Analyze the risks. Describe, in this context, what a Type I error and a Type II error would each mean for the club, and say which one you would most want to avoid here — a judgment the lesson's tables do not make for you.

3 · Decide and interpret. Suppose the test returns a p-value of 0.04. State your decision against the significance level you chose in step 1 (note it would differ at 0.05 versus 0.01). Then, if the result is significant, explain why even a significant result might still not justify switching every class to the tool. **In pairs,** compare interpretations and agree on what evidence beyond the p-value you would still want.

Give evidence for your answer.

Stuck? Start from what you are trying to disprove: the null is always “no difference.” Then ask what it would cost to reject it wrongly (Type I) versus to miss a real effect (Type II) — that cost is what should guide your significance level.

🌍 In a New Context

A shop tests two checkout-button colors with data from 200,000 visitors and finds the new color's click rate is 0.05 percentage points higher, with a p-value of 0.01 at the 5% significance level. State whether the result is statistically significant, and explain why the shop might still decide not to change the button.

❓ Lesson Question — Answered

The lesson asked how we reason from a sample to a whole population and how we measure the uncertainty in that reasoning. Because a sample is only part of the population, we use statistical inference — either estimation, which reports the population as a single value (point estimation) or as a range (interval estimation) whose confidence level says how often such intervals would contain the true value, or hypothesis testing, which sets a null hypothesis ("no difference") against an alternative and compares a p-value with a chosen significance level to decide whether to reject the null. The uncertainty is never removed: it is measured and reported. Judgment can fail as a Type I error (rejecting a true null) or a Type II error (keeping a false null), and lowering one raises the other. And a result that is statistically significant — especially with very large samples — is not automatically important in practice, so results must be interpreted, not just computed.

Exercise

1 Cover the Point! section with a red plastic sheet and quiz yourself in your notebook in numerical order.

2 Read the following passage and answer each question.

In statistical inference, the properties of the population are inferred from the sample. Estimating the properties of the population as a single value is called (a), and estimating within a certain range is called (b). Also, the procedure of judging a hypothesis about the population is called (c). In hypothesis testing, the (d)—such as "no difference" or "no relationship"—and the opposite (e) are set up, and a judgment is made by comparing the (f) computed from the data at hand with the predetermined (g).

1. Fill in the blanks (a)–(g).

⭐ REFLECT & CHALLENGE

Reflect: Why is a 95% confidence level a statement about the method, not about one particular interval?
Challenge: A study of 500,000 users reports a "highly significant" effect ($p < 0.001$) whose size is 0.02 percentage points. Argue whether you would act on it, and name the one extra piece of information that would change your mind.

3 Answer the following questions.

(1) From the following options (A–D), choose the one that most appropriately describes confidence intervals and confidence levels.

A A confidence interval with a 95% confidence level means that the true population value is contained in that interval with a probability of 95%.

B The higher the confidence level, the narrower the confidence interval.

C A confidence interval with a 95% confidence level means that, if the same survey method is repeated, about 95% of the resulting confidence intervals will contain the true population value.

D The confidence level is a value used as the criterion for judgment in hypothesis testing.

(2) From the following options (A–D), choose the one that is NOT an appropriate description of errors in hypothesis testing.

A A Type I error is rejecting the null hypothesis when it is actually correct.

B A Type II error is not rejecting the null hypothesis when it is actually incorrect.

C Lowering the significance level can simultaneously reduce both Type I and Type II errors.

D Type I and Type II errors have a trade-off relationship.

(3) From the following options (A–D), choose the one that most appropriately describes the interpretation of statistical inference.

A A statistically significant difference is necessarily an important difference in practice.

B With a large amount of data, even a tiny difference can become statistically significant.

C If the p-value falls below 0.05, the conclusion is certainly correct.

D Statistical inference can completely remove uncertainty.

⭐ Key takeaway

Remember:
Estimate with a point or an interval (confidence level = how often the method captures the truth); test by comparing a p-value with a significance level to reject or keep the null. Watch Type I / II errors, and never confuse statistical significance with practical importance.